

Physics-0

Introduction

Course Information

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Email: wangqr@mail.tsinghua.edu.cn

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Office: 双清综合楼A座B522

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- **Teaching assistant:**

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宝音夫, byf23@mails.tsinghua.edu.cn

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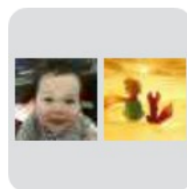
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陈默, will send alerts to students who have missed homework submissions or have been absent from classes



群聊：物理0 2025春



该二维码7天内(2月24日前)有效，重新进入将更新

Goal

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- Building a **global picture** about physics

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- Building a **global picture** about physics
- Getting more familiar with the way of **thinking** in physics

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- Getting more familiar with the way of **thinking** in physics
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- Q: What level of physics knowledge do you think you have?

Topics

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- Newtonian dynamics

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- Newtonian dynamics
- Electromagnetism and electrodynamics

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- Newtonian dynamics
- Electromagnetism and electrodynamics
- A visit to Tsinghua University Science Museum
(?)

References

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1. Lecture notes on 网络学堂

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2. D. Halliday, R. Resnick, J. Walker, Fundamentals of Physics 《基础物理学》

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The latter two are too long to follow...

Gratings

Gradings

- Only homework and final exam
- The grading scheme will be decided by Qiuzhen admission office
- 作业：每周二课后在网络学堂布置作业，下周二24:00前提交到网络学堂（需拍照合并为一个pdf文件上传），助教网上批阅。

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- 高中：知道焦耳定律，所以会上课玩手机取暖，并且能完美躲开老师扔过来的粉笔头抛物线轨迹

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- 爱因斯坦：私下会用引力波给三体文明发朋友圈

What is Physics

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- A science that studies **nature**.

What is Physics

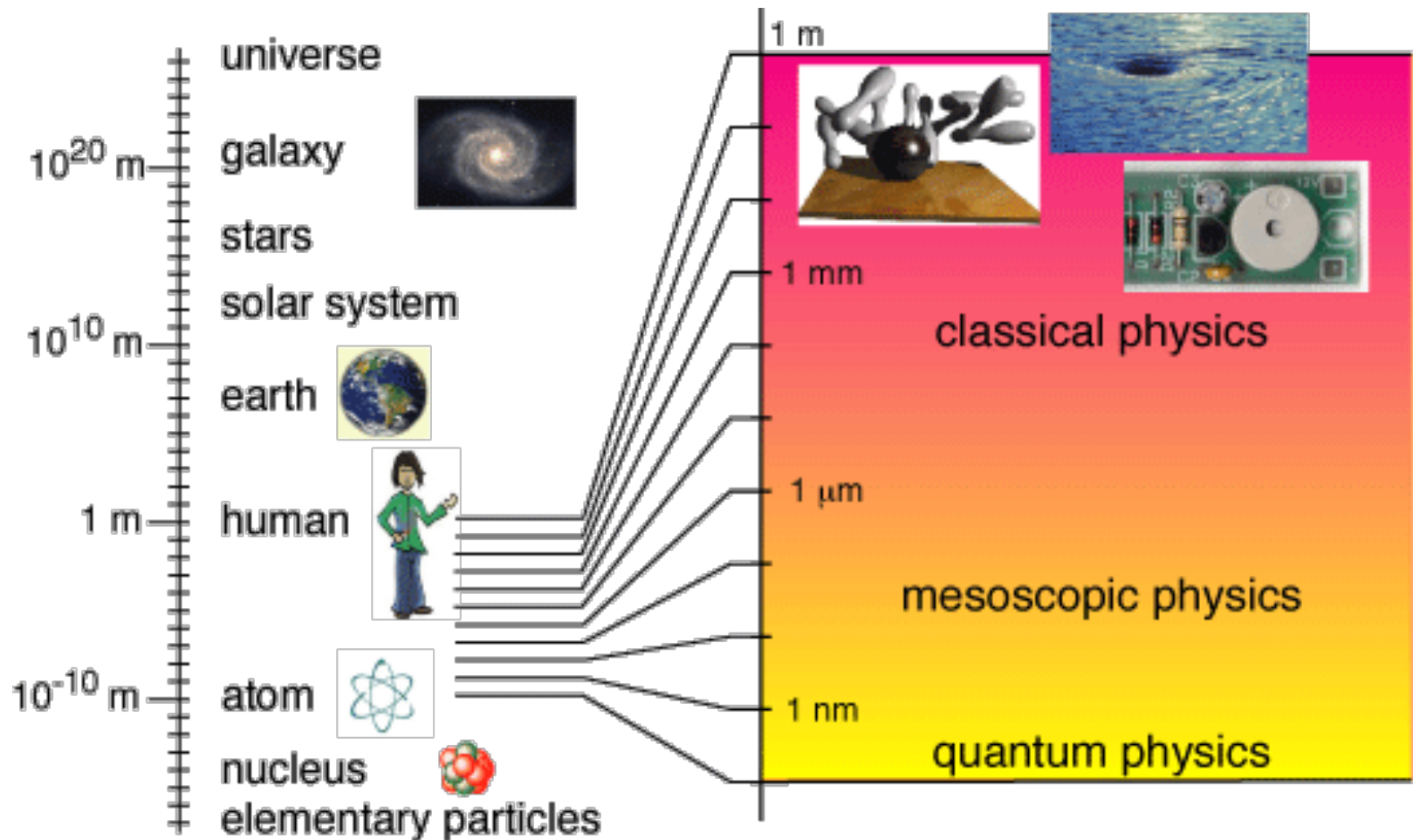
- A science that studies **nature**.
- To describe and understand all natural phenomena

What is Physics

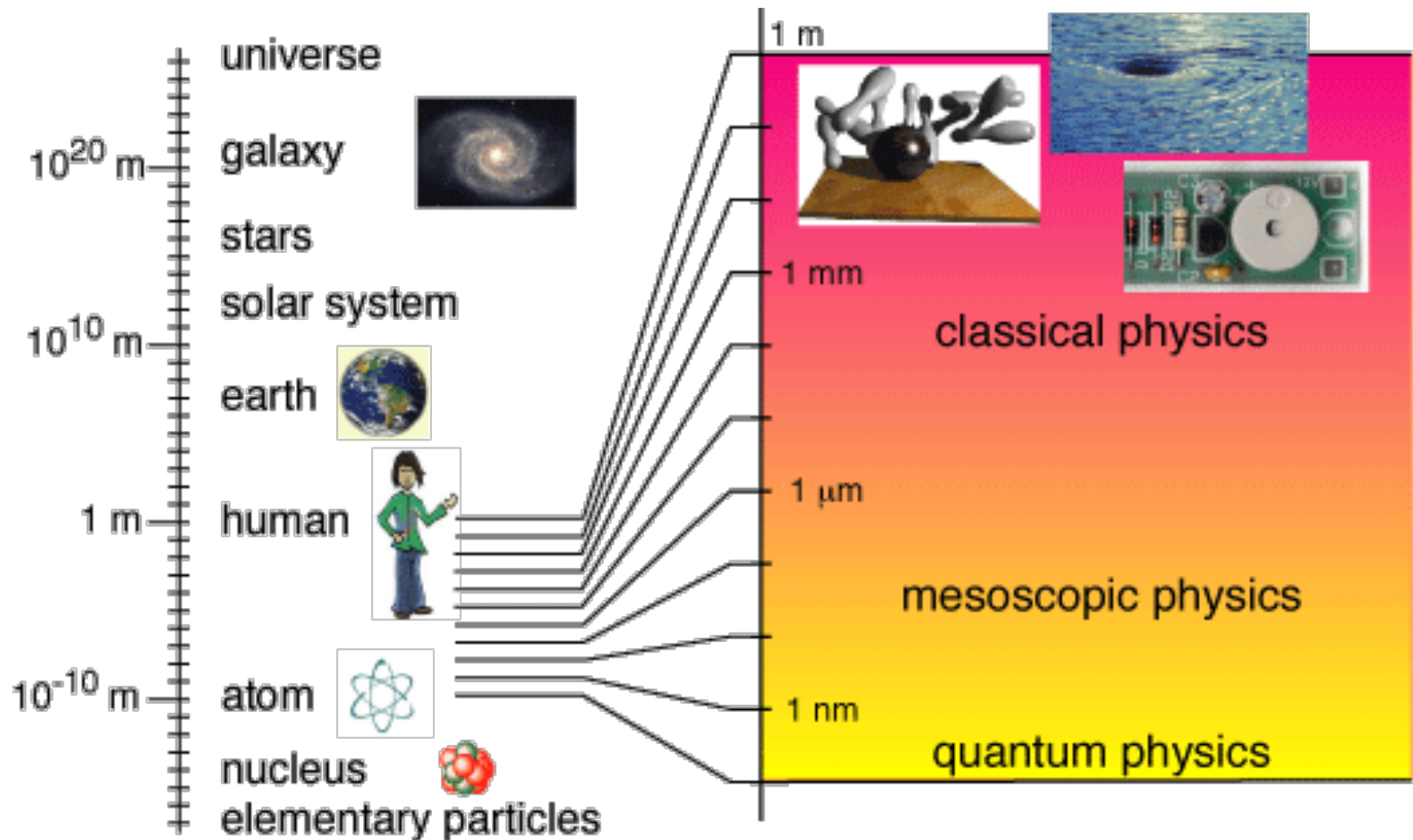
- A science that studies **nature**.
- To describe and understand all natural phenomena
- “Physical science is that department of knowledge which relates to the order of nature, or, in other words, to the regular succession of events.” (Maxwell 1878)

Length scale of physics

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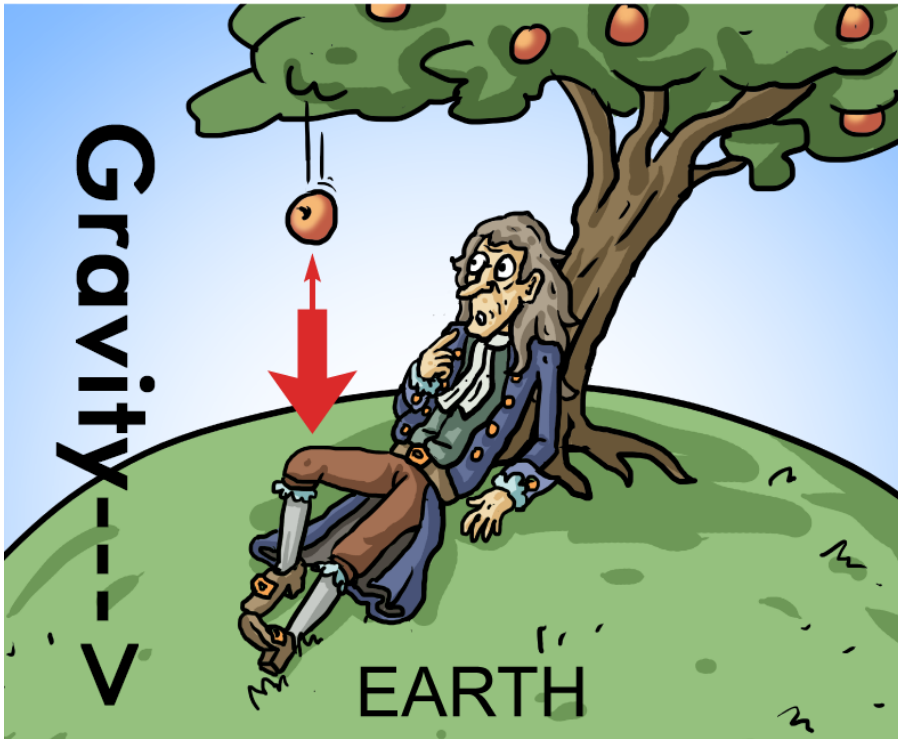
Length scale of physics



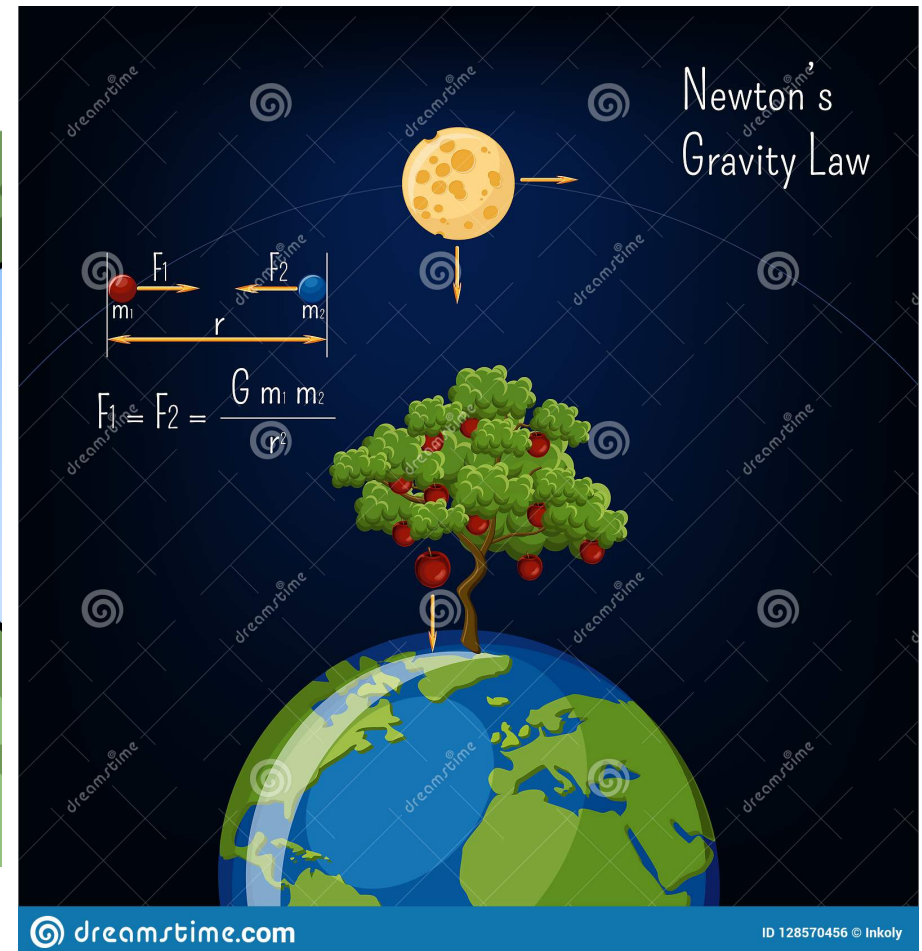
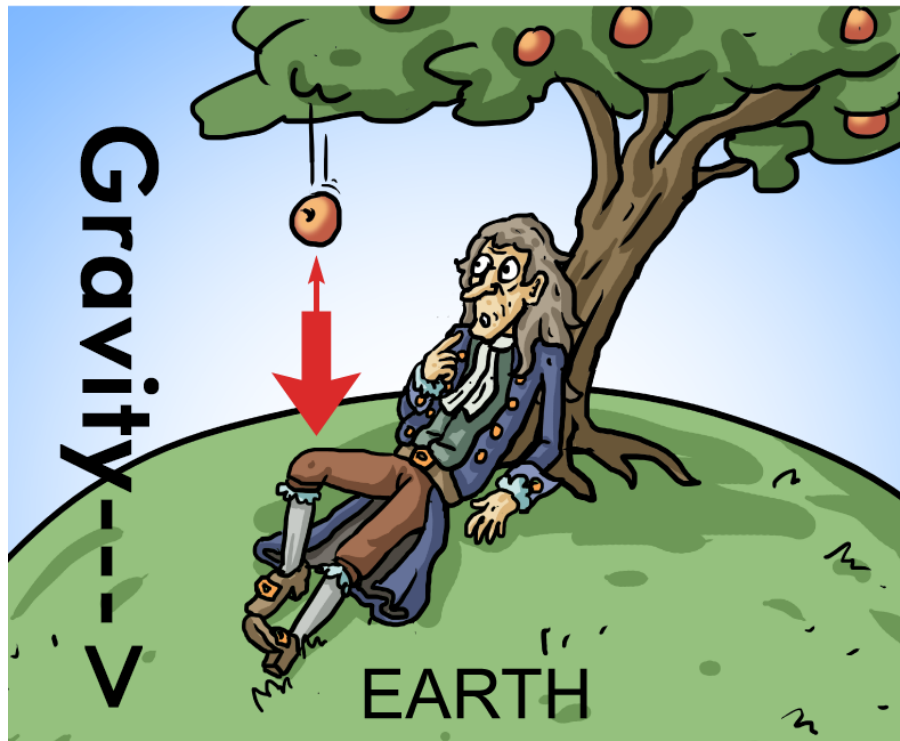
Planck length $\sim 10^{-35}$ m

Revolution = unification in physics

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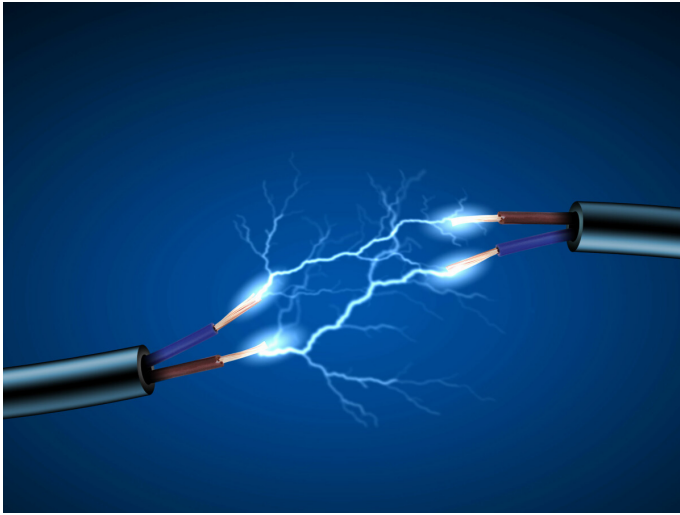


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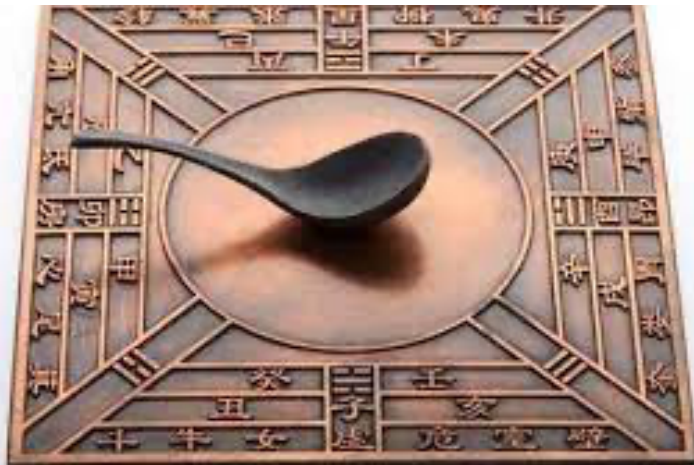
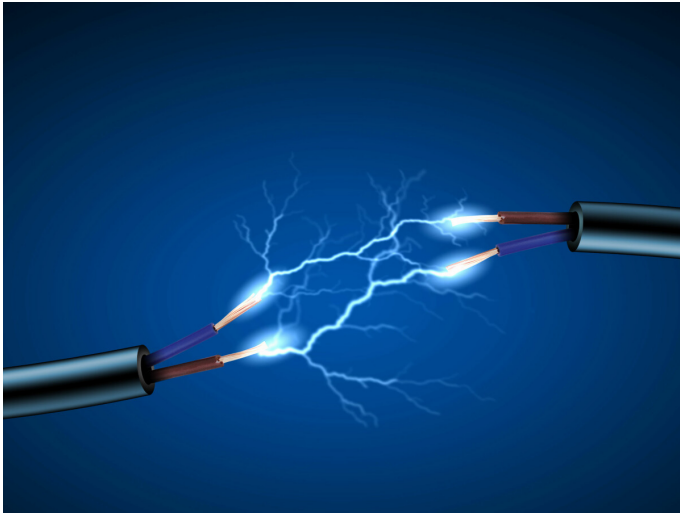


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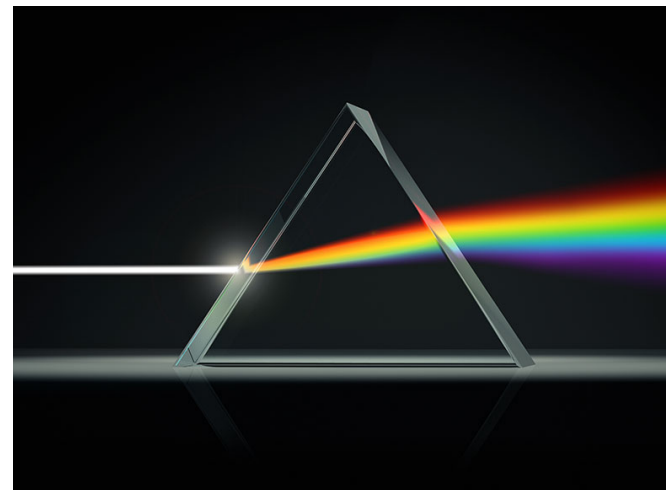
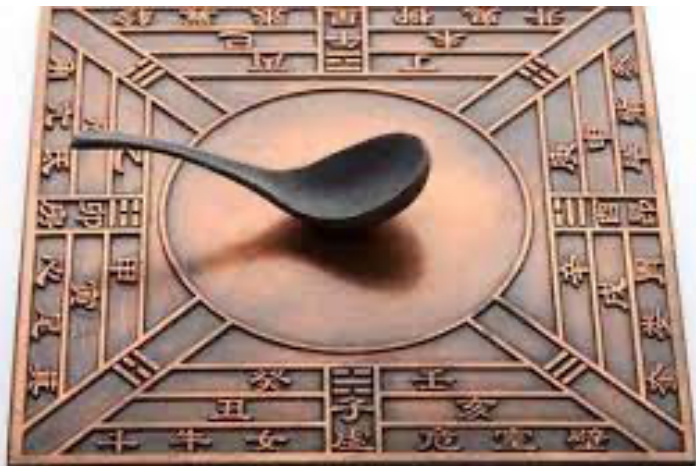
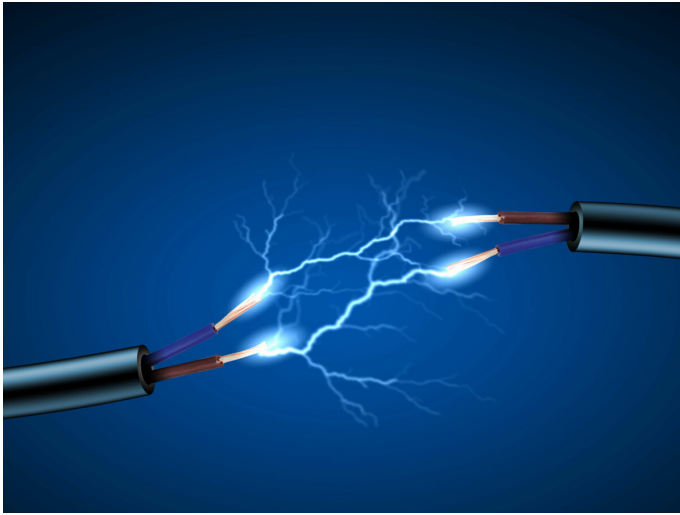
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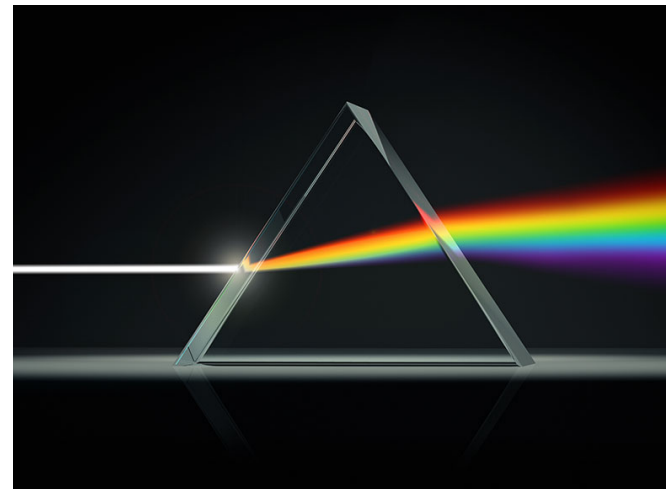
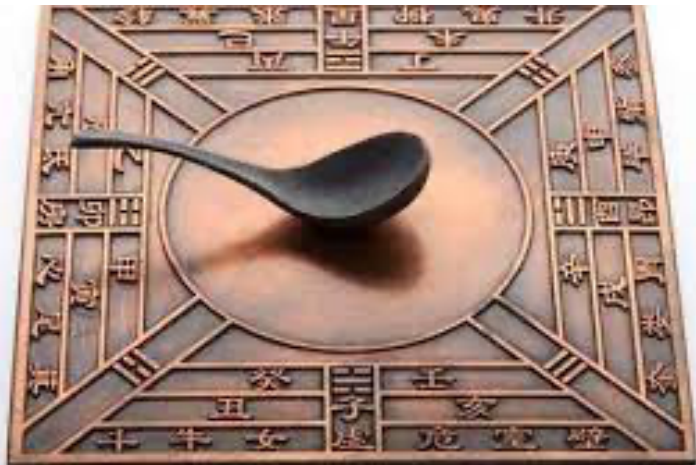
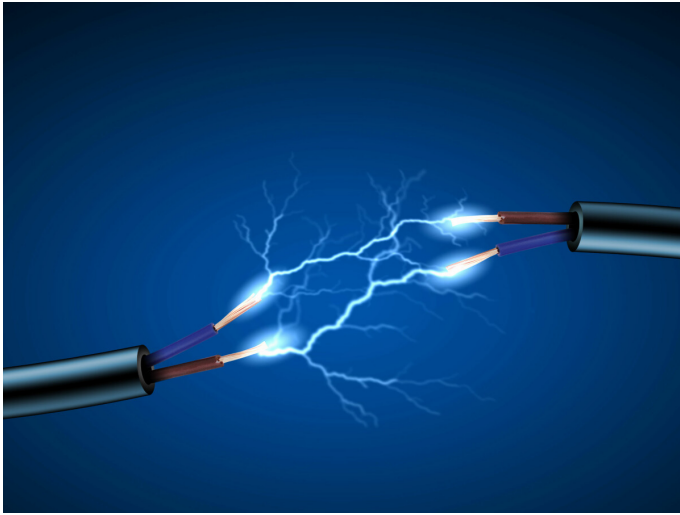
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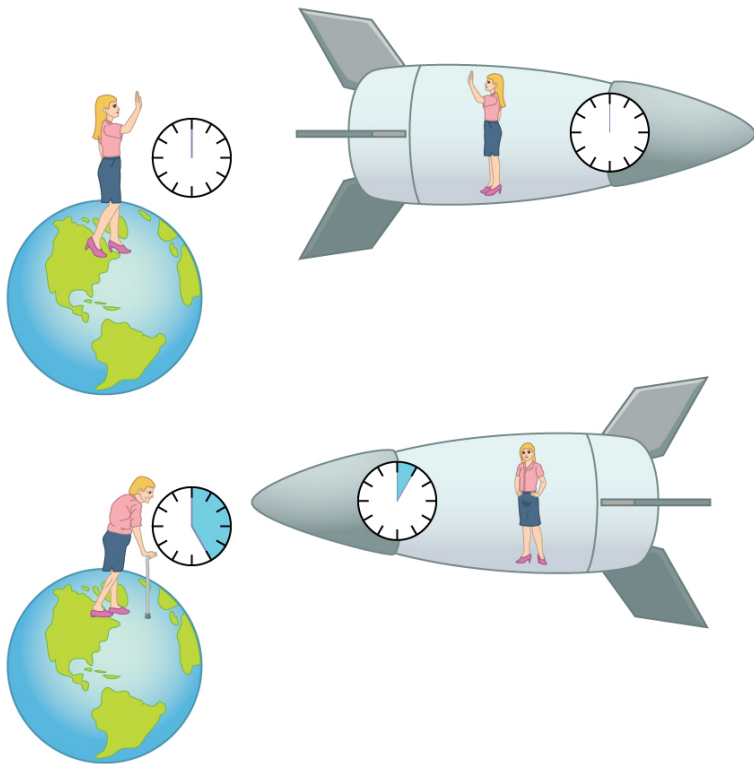


Revolution = unification in physics

- Special and general relativity

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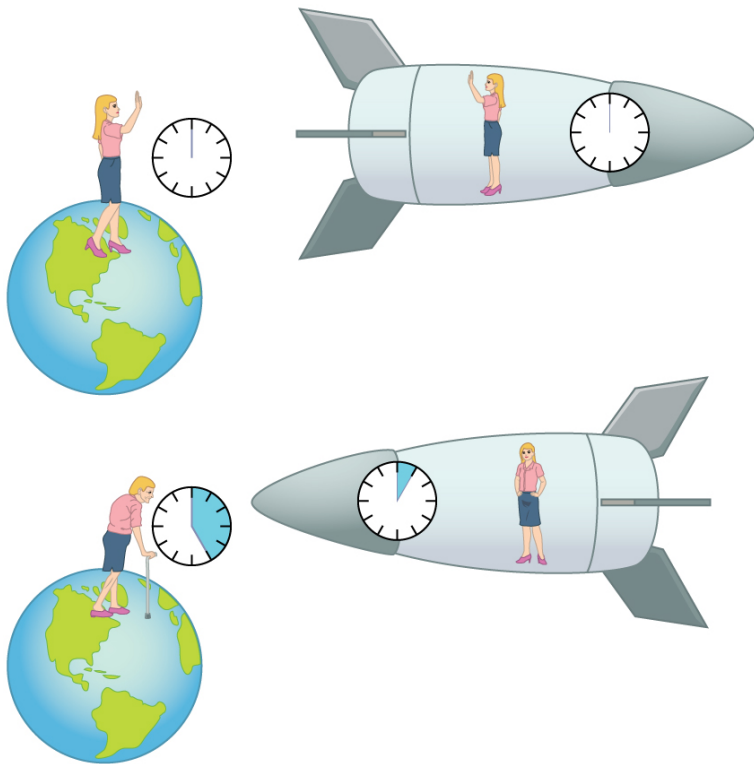
- Special and general relativity



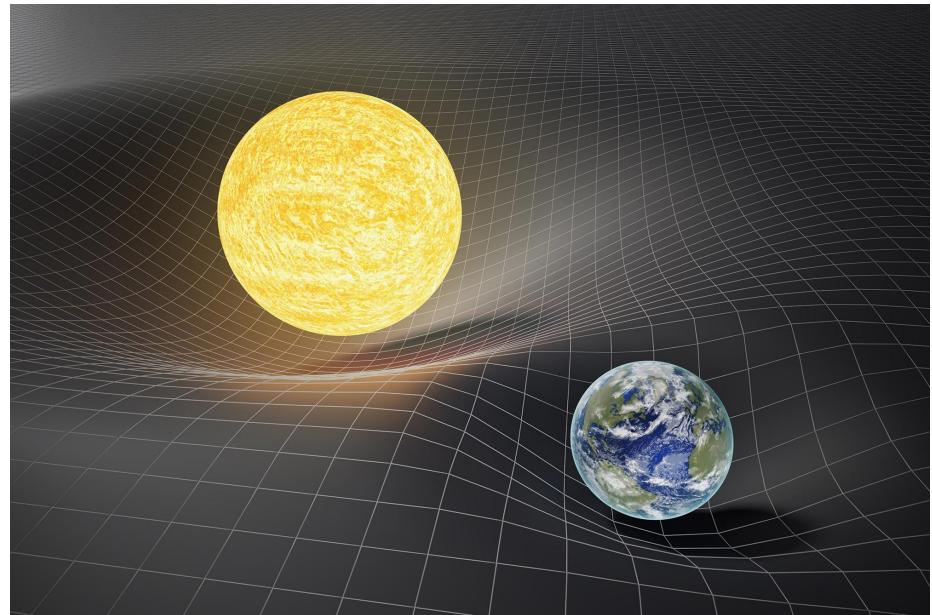
spacetime

Revolution = unification in physics

- Special and general relativity



spacetime



gravity = geometry

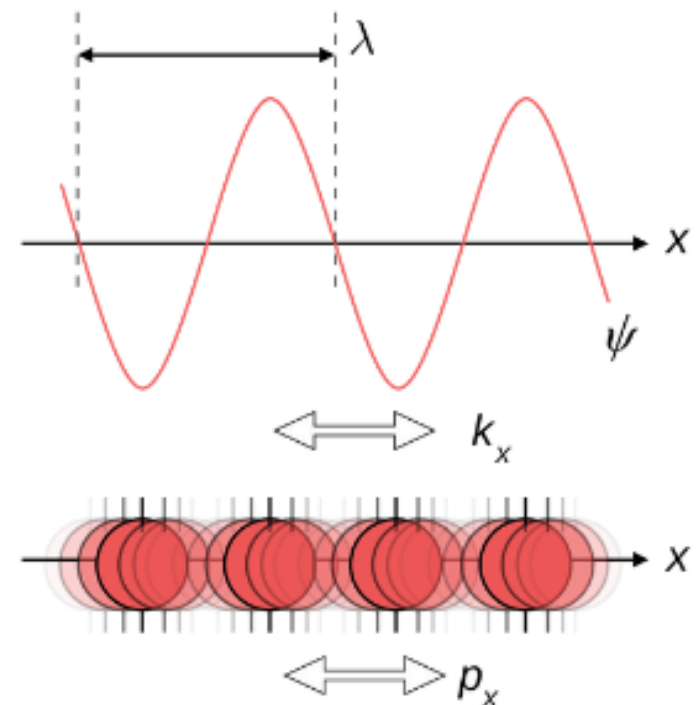
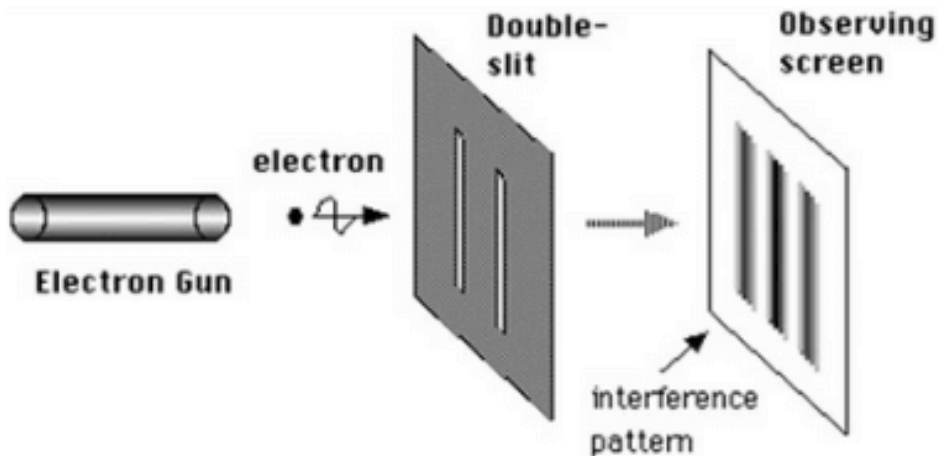
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- Quantum physics

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What is Wave Particle Duality?

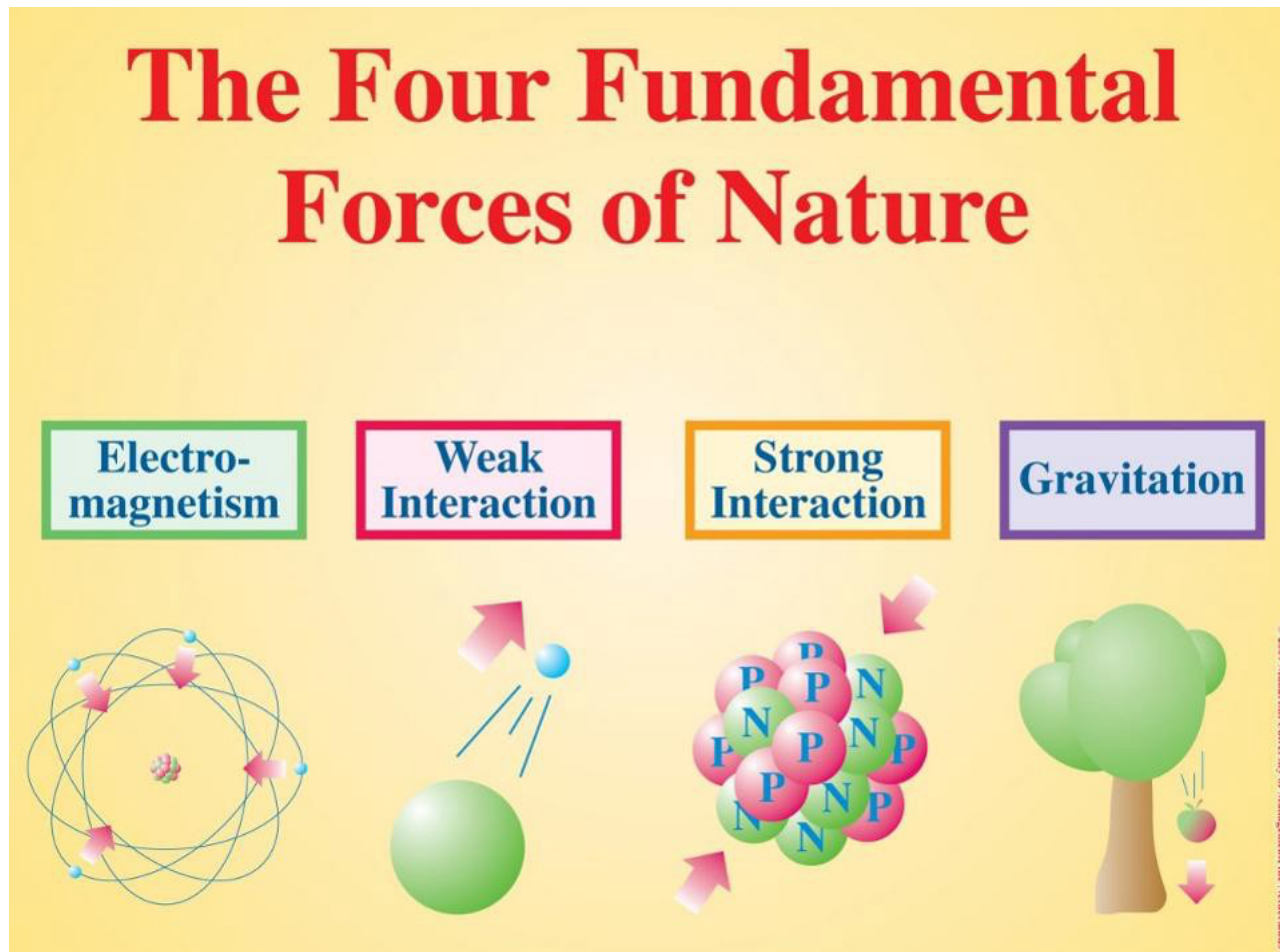


Revolution = unification in physics

- Gauge theory

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- Gauge theory



Relation to Mathematics

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Three examples:

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- Newtonian mechanics and calculus

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- Einstein's general relativity and Riemannian geometry

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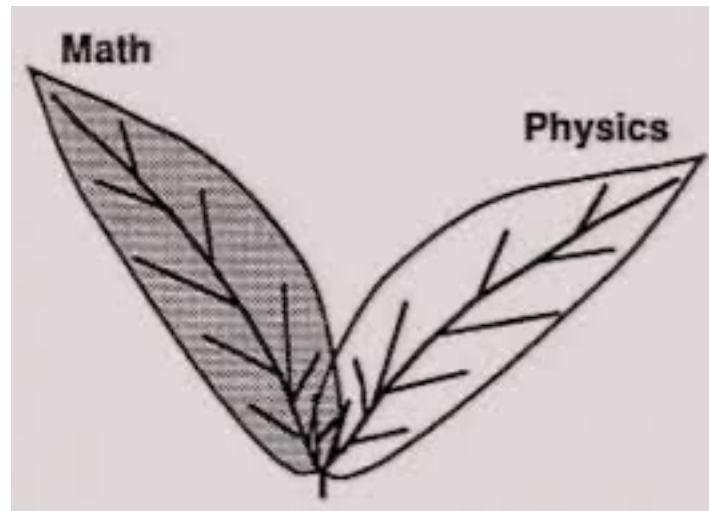
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- Quantum field theory and (?)

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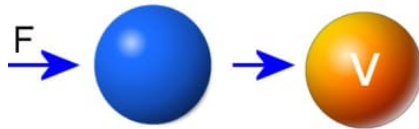
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- ...

1. Newtonian dynamics

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Newton's Laws of Motion

1st Law



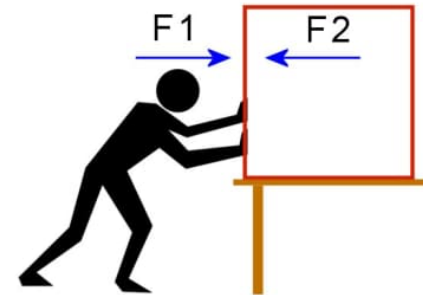
v forever

2nd Law



$$F = ma$$

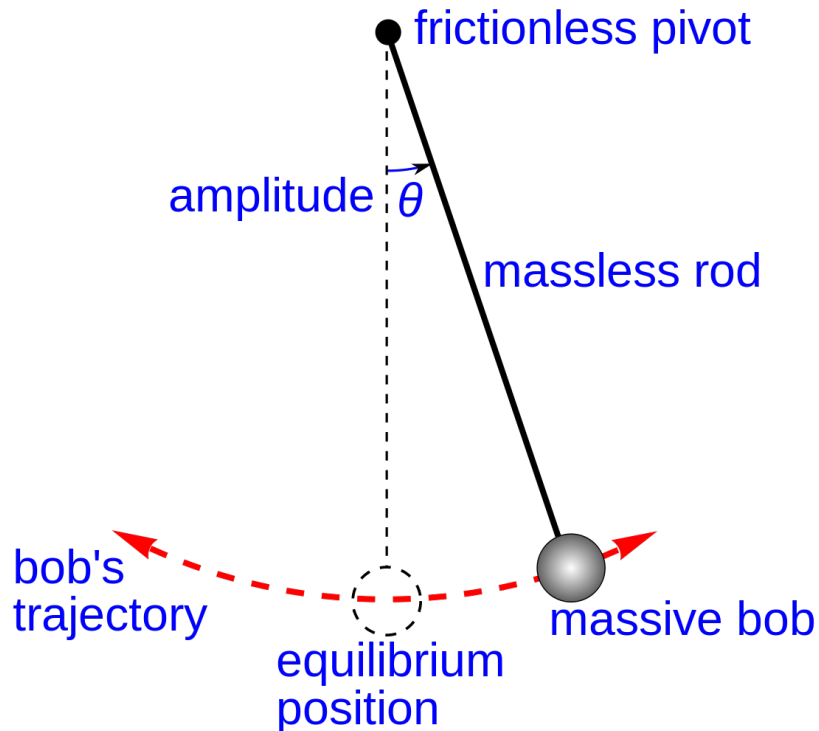
3rd Law



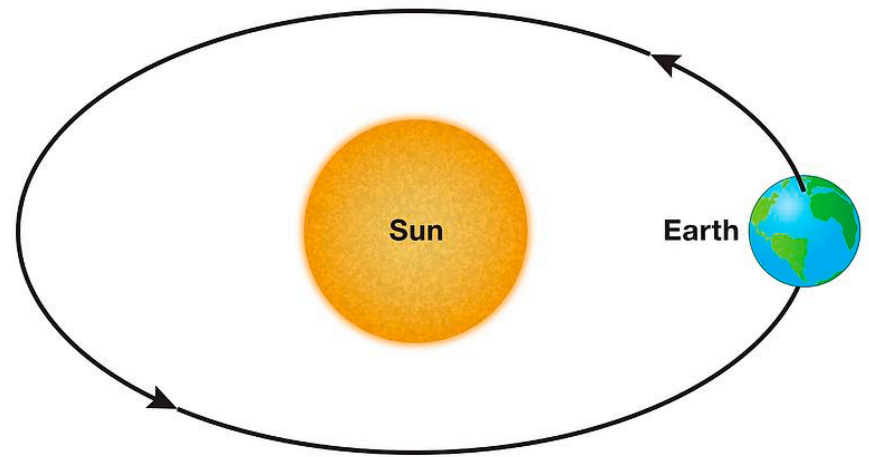
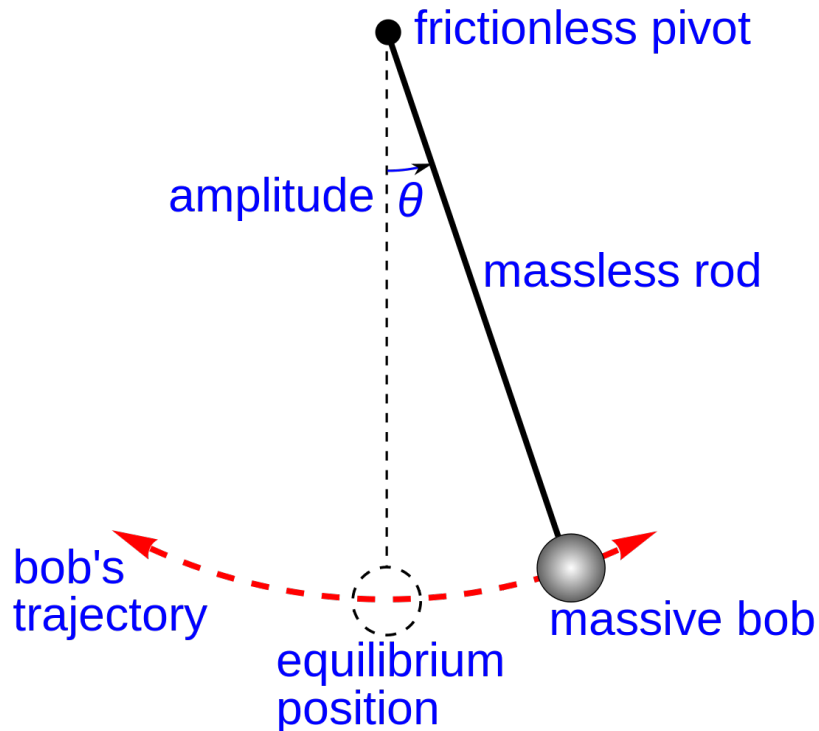
$$F_1 = F_2$$

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2. Electromagnetism and electrodynamics

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$$\nabla \cdot \mathbf{D} = \rho \quad (1) \quad \text{Gauss' Law}$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2) \quad \text{Gauss' Law for magnetism}$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t} \quad (3) \quad \text{Faraday's Law}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} \quad (4) \quad \text{Ampère-Maxwell Law}$$

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$$\begin{aligned} dF &= 0 \\ d^* F &= J \end{aligned}$$

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